Notes 2/19

Thm 11) I know h^{-1} is 1:1

If $a \in S$, define $h^{-1}(a)$ to be the $k \in \{1, \dots, n\}$ such that $h(k) = a$.

Since h is onto $\forall\, a \in S\ \exists\, k \in \{1, \dots n\}$ such that $h(k) = a$

Second, since h is injective, if $h(k) = h(l) = q$

then $k = l$.

3.3: A new perspective on addition

Alternative Definition. Let $m, n \in \mathbb{N}$. Define $m + n \overset{\text{def}}{=} |A \cup B|$

where A and B are disjoint sets with $|A| = m$ & $|B| = n$

Associativity: $A \cup B = B \cup A$

Thm 22: If $A \subseteq C$, then $|A| \leq |C|$. If $A \subsetneq C$, then $|A| < |C|$

Proof sketch: Consider $C - A = C \setminus A$

Let $b = |C - A|$. If we let $|A| = a$ & $|C| = c$

then we see $C = A \cup (A - C)$, so $|C| = |A \cup (C - A)|$.

disjoint $\sqcup$

Thus $c = a + b$

By Ch. 2 $a \leq c$ with equality if and only if $b = 0$

Thm 23. Let $f: A \to B$ be injective.

Suppose $|B| = b < \infty$ Then the size of $|A| = a < \infty$

and $a \leq b$. If f is not surjective then $a < b$

Proof: Consider $f(A) = \{b \in B;\ f(a) = b \text{ for some } a \in A\} \subseteq B$

Let $g: A \to f(A)$ given by $g(a) = f(a)$. Note g is injective and surjective (by def of $f(A)$) g is bijective. By Thm 10 $|A| = |f(A)| = a$

Applying Thm 22 to $f(A) \subseteq B$, we see $a = |A| \leq |B| = b$. If f is not surjective, then $f(A) \subsetneq B$ and Thm 22 says $a < b$.

3.5: Multiplication!

The Cartesian product of two sets A and B is $A \times B \overset{\text{def}}{=} \{(a,b;\ a \in A,\ b \in B\}$

x-y plane is $\mathbb{R} \times \mathbb{R}$

For $m, n \in \mathbb{N}$ define $m \times n = m \cdot n = mn \overset{\text{def}}{=} |A \times B|$ where $|A| = m$ and $|B| = n$

$2 \times 3 = |A \times B|$ where $A = \{0, \square\}$ and $B = \{\star, \Delta, \Diamond\}$

$A \times B = \{(0,\star)(0,\Delta)(0,\Diamond)(\square,\star)(\square,\Delta)(\square,\Diamond)\}$ Six elements

3.7 A new perspective on Exponentiation

Def. For sets A and B define B^A to be the set of functions from A to B.

Exponentiation: if $|A| = a$ and $|B| = b$, then $|B^A| = b^a$

Example: $A = \{0, \square\}$ $B = (\star, \Delta, \Diamond)$ 9 different functions

3^2

if $|A| > |B| \Rightarrow f: A \to B$ is NOT injective (pigeon hole)

Ch4. The Integers: How to define integers

We will do it by thinking of differences between sets

Say B has size 4 and A has size 3. $(3,4) \approx -1$

Def: Let $\Delta: \mathbb{N} \times \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ be defined by $\Delta(m,n) = \begin{cases} (m-n, 0) & \text{if } m > n \\ (0, n-m) & \text{if } n > m \\ (0,0) & \text{if } n = m \end{cases}$

$(0, n-m)$ is the negative case.

This is well defined by trichotomy: $\Delta(4,3) = (1,0) =$ "+1"

$\Delta(3,4) = (0,1) =$ "-1"

$\Delta(3,3) = (0,0) =$ "0"